

Software Test Plan (STP)

Collaborative Study Material & Quiz Platform

Project: Collaborative Study Material & Quiz Platform

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Collaborative Study Material & Quiz Platform v1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing the platform against the Software Requirements Specification (SRS) v1.0.

Scope: Testing covers user registration and authentication; study material upload, search, versioning and plagiarism screening; quiz creation and management; quiz taking with auto-grading; discussion and rating; leaderboards, analytics and notifications; and administration and moderation. Non-functional testing covers performance, availability, scalability, audit logging, accessibility (WCAG 2.1 AA) and security (CSQ-SR-001 to CSQ-SR-005).

References: Software Requirements Specification (SRS) v1.0 – Collaborative Study Material & Quiz Platform; Software Architecture and Design Specification (SAD) v1.0; WCAG 2.1 AA; OWASP ASVS; IEEE 829 (test documentation).

Definitions: SRS – Software Requirements Specification; SAD – Software Architecture and Design; STP – Software Test Plan; RTM – Requirements Traceability Matrix; RBAC – Role-Based Access Control; JWT – JSON Web Token; TLS – Transport Layer Security; UAT – User Acceptance Testing; WCAG – Web Content Accessibility Guidelines; TC – Test Case.

2. Test Items

- Authentication module (registration, login, logout)
- Study material management module (upload, search, versioning, plagiarism screening)
- Quiz creation & management module
- Quiz taking & auto-grading module
- Discussion & rating module
- Leaderboard, analytics & notifications module
- Admin & moderation module

3. Features to be Tested

Features mapped to SRS requirement IDs:

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- CSQ-F-001: User registration with email, password and role; verification email
- CSQ-F-002: Login via Authentication Service and JWT issuance
- CSQ-F-003: Account lockout for 15 minutes after 5 failed logins; audit event

- CSQ-F-010: Upload of study material with subject, topic and difficulty tags
- CSQ-F-011: Plagiarism scan before publishing; flagging above threshold
- CSQ-F-012: Search and filter of material (results within 2 s for 10,000 items)
- CSQ-F-013: Version history and download of previous versions
- CSQ-F-020 / 021: Quiz creation from question bank; quiz settings (time limit, attempts, shuffle, difficulty)
- CSQ-F-022 / 023: Quiz publishing to groups with availability window; question reuse and editing
- CSQ-F-030: Timed quiz attempt with attempt-count enforcement
- CSQ-F-031 / 032: Auto-grading within 2 s; answer and explanation reveal setting
- CSQ-F-040 / 041 / 042: Comments, threaded replies (3 levels), 5-point material rating
- CSQ-F-050 / 051: Per-quiz leaderboard; instructor dashboard with CSV export
- CSQ-F-052: Notifications for new material, quiz deadlines (24 h prior) and results
- CSQ-F-060 / 061 / 062: Suspend/reactivate users; moderation queue; content removal and warnings
- CSQ-NF-001: Page load $p95 \leq 3$ s under normal load
- CSQ-NF-002: 99.5% monthly availability
- CSQ-NF-003: 5,000 concurrent quiz attempts
- CSQ-NF-004: Activity logging and 2-year retention
- CSQ-NF-005: WCAG 2.1 AA accessibility
- CSQ-SR-001 to 005: TLS 1.2+, password hashing, quiz-content encryption, RBAC, administrative audit logging

4. Features Not to be Tested

- Internal logic of the third-party Plagiarism Detection Service (assumed tested by the vendor); only its integration contract is tested
- Internal logic of the external Authentication Service and the Email/Push notification gateway (vendor responsibility)
- Underlying cloud provider infrastructure reliability (compute, storage, network)
- Features excluded by SRS scope: offline/desktop clients, payment processing, and LMS integration beyond optional single sign-on

5. Test Approach / Strategy

Levels:

- Unit tests (service/module-level)
- Integration tests (service ↔ service, service ↔ database/storage, service ↔ third-party API)
- System tests (end-to-end platform flows, e.g. upload → scan → publish; take quiz → grade → leaderboard)
- Acceptance tests (UAT with sample students and instructors)

Types:

- Functional testing (core features above)
- Regression testing (on every release candidate)
- Performance testing (response time, concurrent quiz load)
- Usability & accessibility testing (UI clarity, WCAG 2.1 AA conformance)

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password handling (hashing, no plaintext logging) — CSQ-SR-002

- TLS 1.2+ verification across all client-server and service-to-service calls — CSQ-SR-001
- RBAC test matrix confirming students cannot access instructor/admin functions — CSQ-SR-004
- Fuzzing of input fields (registration, material upload, quiz answers)
- Penetration testing of authentication and quiz-submission flows

6. Test Environment

Hardware: standard developer/test workstations; mobile devices (Android/iOS) for responsive and mobile client testing.

Software: platform build v1.0, staging API environment, sandboxed Plagiarism Detection and Notification service stubs.

Tools: Selenium / Playwright (UI automation), Postman (API testing), JMeter / k6 (performance and load testing), Jira (defect tracking).

Test Data: dummy student/instructor accounts, sample study material files, seeded question banks.

7. Test Schedule

Milestones:

- Test case design: 10-Oct-2026
- Environment setup: 12-Oct-2026
- Test execution start: 13-Oct-2026
- Test execution end: 24-Oct-2026
- UAT: 27-Oct-2026 to 30-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility	Scrum role
Test Engineer (Integration Testing)	Manasi	Design & execute integration test cases across modules; log defects	Developer
Developer (Frontend & CI/CD)	Mahesh	Maintain CI/CD pipeline; support defect fixes and triage	Developer

Role	Name	Responsibility	Scrum role
Product Owner	Mokshith	Approve test results and acceptance criteria; sign-off readiness	Product Owner
QA Coordinator	Rakshitha	Coordinate test schedule with sprint activities; track execution status	Scrum Master

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the dev team
Test environment downtime	Maintain a backup environment on a cloud VM
Plagiarism Detection / Notification service sandbox instability	Use local stubs/mocks to keep test execution unblocked

11. Assumptions & Dependencies

- Plagiarism Detection and Notification sandbox APIs will be stable and available during test execution
- Test data (sample accounts, material, question banks) will be provided before execution begins
- Staging environment mirrors production configuration closely enough for performance testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of planned test cases)

Resume testing if:

- Blocking defects are resolved
- Environment is stabilised

13. Test Case Management & Traceability

Req ID	Requirement	Test case(s)
CSQ-F-001	User registration	TC-Auth-01
CSQ-F-002	Login and JWT issuance	TC-Auth-02
CSQ-F-003	Account logout	TC-Auth-03
CSQ-F-010	Upload study material	TC-Mat-01
CSQ-F-011	Plagiarism screening	TC-Mat-02
CSQ-F-012	Search and filter	TC-Mat-03
CSQ-F-013	Material versioning	TC-Mat-04
CSQ-F-020	Create quiz from question bank	TC-QuizC-01
CSQ-F-021	Quiz settings	TC-QuizC-02

Req ID	Requirement	Test case(s)
CSQ-F-022	Publish to groups	TC-QuizC-03
CSQ-F-023	Reuse questions	TC-QuizC-04
CSQ-F-030	Attempt quiz (timed)	TC-Quiz-01
CSQ-F-031	Auto-grading	TC-Quiz-02
CSQ-F-032	Answer reveal	TC-Quiz-03
CSQ-F-040	Comments	TC-Disc-01
CSQ-F-041	Threaded replies	TC-Disc-02
CSQ-F-042	Material rating	TC-Disc-03
CSQ-F-050	Leaderboard	TC-Lead-01
CSQ-F-051	Instructor dashboard	TC-Lead-02
CSQ-F-052	Notifications	TC-Notif-01
CSQ-F-060	Suspend / reactivate user	TC-Admin-01
CSQ-F-061	Moderation queue	TC-Admin-02
CSQ-F-062	Content removal and warning	TC-Admin-03
CSQ-NF-001	Page load $p95 \leq 3$ s	TC-Perf-01
CSQ-NF-002	99.5% availability	Ops uptime reports
CSQ-NF-003	5,000 concurrent attempts	TC-Perf-02
CSQ-NF-004	Audit logging and 2-year retention	TC-OPS-01
CSQ-NF-005	WCAG 2.1 AA	TC-UX-01
CSQ-SR-001	TLS 1.2+	TC-Sec-01
CSQ-SR-002	Password hashing	TC-Sec-02
CSQ-SR-003	Quiz content encryption	TC-Sec-03
CSQ-SR-004	RBAC	TC-Sec-04
CSQ-SR-005	Admin audit logging	TC-Sec-05

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Manasi	Manasi / 05/10/2026
Dev Lead	Mahesh	Mahesh 05/10/2026
Product Owner	Mokshith	Mokshith 05/10/2026
Scrum Master	Rakshitha	Rakshitha 05/10/2026